

March 2023

STRENGTH OF MATERIALS

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A is compulsory and to be answered first. This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided. On early submission of answer scripts of Question No. 1, a student will get the remaining script earlier.

Answer any Five (05) Questions from Group B.

Group A

1. Choose the correct answer from the given alternatives (any twenty): 1x20
- i) The material which shows little or no plastic deformation is called – (a) ductile material (b) brittle material (c) malleable material (d) plastic material.
 - ii) The point at which there is increase in strain with no appreciable increase in stress is known as – (a) Breaking point (b) Yield point (c) Elastic limit (d) Proportional limit.
 - iii) Ratio of Hoop stress and longitudinal stress of a material is equal to – (a) 1 (b) $\frac{1}{2}$ (c) 2 (d) none of these.
 - iv) Temperature stress developed in a bar depends upon – (a) co-efficient of linear expansion (b) change of temperature (c) Young's modulus (d) all of these.
 - v) The area moment of inertia is often called _____ moment of area. – (a) first (b) second (c) third (d) fourth
 - vi) Moment of Inertia of a triangular lamina of height 'h' about its base 'b' is – (a) $\frac{b \times h^3}{12}$ (b) $\frac{b \times h^3}{36}$ (c) $\frac{h \times b^3}{12}$ (d) none of these.
 - vii) If I_{OX} and I_{OY} be moments of inertia of a lamina about mutually perpendicular axes OX and OY in the plane of the lamina and I_{OZ} be the moment of inertia of the lamina about an axis OZ normal to the lamina and passing through the point of intersection of the axis OX and OY then $I_{OZ} = I_{OX} + I_{OY}$. The above theorem is known as – (a) theorem of parallel axes (b) theorem of perpendicular axes (c) three moments theorem (d) none of these.
 - viii) A simple supported beam of length 10 m carries uniformly distributed load of 10 kN/m over the entire span. The magnitude of each support reaction will be – (a) 25 kN (b) 30 kN (c) 50 kN (d) 100 kN.
 - ix) The maximum bending moment of a cantilever beam appears at – (a) fixed end (b) free end (c) mid-span (d) none of these.
 - x) If a beam is loaded with UDL, the nature of bending moment diagram will be – (a) straight horizontal line (b) straight inclined line (c) parabolic (d) cubic.
 - xi) At the point of contra flexure, the value of bending moment will be – (a) maximum (b) minimum (c) zero (d) none of these.
 - xii) The bending stress at neutral axis is – (a) maximum (b) minimum (c) zero (d) tensile in nature.
 - xiii) The section modulus of a circular section of diameter d is – (a) $\frac{\pi d^2}{32}$ (b) $\frac{\pi d^3}{32}$ (c) $\frac{\pi d^3}{64}$ (d) $\frac{\pi d^4}{64}$.
 - xiv) Neutral axis passes through – (a) the top layer of the beam (b) the bottom layer of the beam (c) the CG of the beam (d) none of these.

- xv) Maximum deflection of a beam occurs where the slope of the beam is – (a) maximum (b) 45° (c) 90° (d) 0° .
- xvi) The maximum deflection of a cantilever beam appears at – (a) fixed end (b) free end (c) mid-span (d) none of these.
- xvii) Doubling the span of beam will result in increase in deflection of the beam by – (a) two times (b) four times (c) eight times (d) none of these.
- xviii) Euler's formula for crippling load is best suitable for slenderness ratio more than – (a) 50 (b) 70 (c) 80 (d) 100
- xix) The crippling load of a column depends on – (a) cross-sectional area (b) modulus of rigidity (c) slenderness ratio (d) all of these.
- xx) The ratio of actual length to the equivalent length of a column having one end fixed and other end free is – (a) 1:1 (b) $\sqrt{2}$:1 (c) 1:2 (d) $1:\sqrt{2}$.
- xxi) The strength of a hollow shaft for the same length, material and weight is _____ a solid shaft. – (a) less than (b) more than (c) equal to (d) none of these.
- xxii) In pure torsion, the shaft gets deformed due to – (a) bending moment only (b) axial load only (c) twisting moment only (d) all of these.
- xxiii) For a hollow circular shaft torsional shear stress at the centre is – (a) maximum (b) zero (c) minimum (d) none of these.
- xxiv) The predominant effect of an axial tensile force on a helical spring is – (a) bending (b) compression (c) twisting (d) tension.
- xxv) Two helical springs of stiffness K is attached in parallel. The stiffness of the equivalent system will be – (a) 2K (b) K/2 (c) 2K (d) 4K

Group B

2.
 - i) Draw the stress-strain diagram of Mild Steel and locate all the salient points in the diagram.
 - ii) A steel wire 2 m long and 3 mm in diameter is extended by 0.75 mm when a weight W is suspended from the wire. If the same weight is suspended from a brass wire, 2.5 m long and 2 mm in diameter, it is elongated by 4.64 mm. Determine the modulus of elasticity of brass if that of steel be $2.0 \times 10^5 \text{ N/mm}^2$. 3+5
3.
 - i) Define volumetric strain and bulk modulus of a material.
 - ii) A copper bar of diameter 25 mm and length 2 m is rigidly fixed at both ends. Find the thermal stress induced in the bar if the temperature is reduced by 40°C . Take $E = 100 \text{ GPa}$, $\alpha = 16 \times 10^{-6}/^\circ\text{C}$. 2+6
4.
 - i) State the "Theorem of parallel axis".
 - ii) Show that the moment of inertia of a circular section with respect to an axis passing through its centre of gravity is $\pi D^4/64$, where D is the diameter of the circular section. 2+6
5.
 - i) Write down the relationship between shear force and bending moment and explain its significance.
 - ii) A simply supported beam 2.5 m long is carrying two point loads of 2 kN and 4 kN at a distance of 1 m & 1.5 m from left end respectively. Draw the SF & BM diagram. 2+6
6.
 - i) What are the assumptions in Simple Bending Theory?
 - ii) A simply supported beam of span 5 m has a cross section 150 mm x 250 mm. If the permissible stress is 10 N/mm^2 , find the maximum intensity of uniformly distributed load it can carry. 2+6
7.
 - i) Write the formula for maximum slope of a cantilever beam with a point load at free end.
 - ii) A cantilever beam of 2 m length having width 100 mm and depth 120 mm is subjected to a point load of 30 kN at its free end. Find the maximum slope and deflection at free end if the Young's Modulus of the beam material is 200 GPa. 1+7

8. i) What is slenderness ratio?
ii) Find the Euler's crippling load for a hollow cylindrical steel column of 38 mm external diameter and 2.5 mm thick. Take length of the column as 2.3 m and hinged at its both ends. Take $E = 205 \text{ GPa}$. 2+6
9. i) Write down the equation of Pure torsion and explain each term.
ii) A solid circular shaft transmits 100 kW power at 200 r.p.m. Calculate the shaft diameter if twist in the shaft is not to exceed 1° in 2 metres of the shaft length and the shearing stress is limited to 70 N/mm^2 . Take modulus of rigidity $= 8 \times 10^4 \text{ N/mm}^2$. 2+6
10. i) What is Spring index?
ii) A close coil helical spring to carry a load of 250 N, if the mean coil diameter is to be 8 times that of the wire, calculate these diameters. Take maximum shear stress as 70 MPa . 2+6
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